Moreno crime network





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1. Introduction
2. Dataset
3. Research questions
4. Methods
5. Result

Introduction

What is a crime network?

A **crime network** is a group of people who work together to commit crimes in an organized way. They are important to study because they help us understand how organized crime actually works.

- They reveal how criminals collaborate
- They help identify key players
- They show the structure of criminal organizations
- They help disrupt and prevent crime
- They support scientific research



Dataset

This bipartite network contains persons who appeared in at least one crime case as either a suspect, a victim, a witness or both a suspect and victim at the same time. A left node represents a person and a right node represents a crime. An edge between two nodes shows that the left node was involved in the crime represented by the right node.

Dataset source: http://konect.cc/networks/moreno_crime

Dataset details: Graph with 1380 nodes and 1476 edges

The network format: Bipartite.

Filtering keeping only the “suspect” edges.

After filtering and the projection: 490 nodes and 945 edges.

```
1 1 {'role': 'Suspect'}
1 2 {'role': 'Victim'}
1 3 {'role': 'Victim'}
1 4 {'role': 'Suspect'}
1 756 {'role': 'Suspect'}
2 5 {'role': 'Victim'}
2 6 {'role': 'Suspect'}
2 7 {'role': 'Victim'}
2 8 {'role': 'Suspect'}
```

Research questions:

1. How can network centrality measures distinguish between different structural roles in a criminal organization?

2. Can we identify lone repeaters, gang leaders, and peripheral 'small fish'?

Bipartite Network: projection on the people

1. Our analysis of the network started with the projection on the people: visually connecting people throughout the participation to a specific crime.
2. From here, the structure of criminal organizations is revealed from the computation of the Centrality measure.
3. In particular the focus will be on: Degree, Closeness, Betweenness and Eigenvector centralities.

Degree Centrality

- **What** is it? Generally speaking, it's a raw (or normalized) computation of the number of connection for each node. In our case, it computes the number of co-offenders one person is connected to through a crime.
- **Why** use it? In our case, a high centrality can be a signal of an individual participating in a large group or gang of offenders, an individual that commits a high number of crimes with few co-offenders, or a mix of both
- **Issue**: Because of the projection of the bipartite network, someone that committed only one crime with a lot of co-offenders will also result in a high degree even though he is not "central".

Degree Centrality

- The real crime repeaters are identified by a high number of crimes and low number of average co-offenders per crime: **815.**
- **95**, **286** and **281** could be both gang members and artefacts for their low number of crimes and high number of average, in this case the criminal is not a strong re-offender as the others score comes from a high average.

Person 18 - 8...	Degree Centrality 0.03476482617...	Num Crimes 17 - 27	Total Co-offenders 51 - 179	Avg Co-offenders per Crime 1.889 - 10.529
815	0.05521472393	27	51	1.889
285	0.04703476483	23	161	7
178	0.04294478528	21	163	7.762
695	0.03885480573	19	138	7.263
715	0.03885480573	19	62	3.263
430	0.03680981595	18	126	7
18	0.03476482618	17	53	3.118
281	0.03476482618	17	139	8.176
286	0.03476482618	17	145	8.529
95	0.03476482618	17	179	10.529

Closeness Centrality

- **What** is it? quantifies how near a person is to all other people in the network. In our case, it measures how much a criminal is near the others (through crimes) and if it's at the center or periphery of the network.

- **Why** use it? In our case, a high centrality can be a signal of an individual at the core of a large group or gang of offenders, someone that can reach others easily and so is well informed about the criminal world.

- **Issue**: In the case of large groups, the projection can collapse into a clique and reduce drastically the distances, thus creating artifacts.

Closeness Centrality

- Again **815** appears at the top, and doesn't look like an artefact. We can now start to think that this person is a real high recidive.

- The members of the top 10 are all possible criminals that have connections with many groups, even though they are not suspects of many crimes, like **43**.

Person int64 43 - 815	Closeness Centr... 0.212508717803...	Num Crimes int64 8 - 27	Total Co-offenders 41 - 179	Avg Co-offender... 1.889 - 14.375
815	0.2335705729	27	51	1.889
695	0.2323894322	19	138	7.263
43	0.2286318917	8	115	14.375
76	0.2238977356	12	117	9.75
409	0.2238977356	12	117	9.75
637	0.2238977356	12	117	9.75
95	0.2193556574	17	179	10.529
592	0.2168968034	14	41	2.929
425	0.2136200359	16	173	10.812
715	0.2125087178	19	62	3.263

Betweenness Centrality

- **What** is it? It ranks nodes based on their control over information flow, meaning the extent to which they lie on paths of other nodes. In our case this can be a measure of connectivity between different groups.
- **Why** use it? Analyzing who serves as a bottleneck between different clusters can be essential for operations: communication relies on this individuals.
- **Issue**: This measure can capture also individuals that appear in giant cliques (they are on all the clique's paths), and so they are not actual bottlenecks.

Betweenness Centrality

- **815** confirms itself at the top, far above the other criminals, meaning that could be an important figure across multiple crimes, even though its average of accomplices is low.
- **95** shows an interesting behavior, as its number of crimes is high and also a high average: maybe the criminal possess information of different crimes and people.
- We still have artefacts like **43** and **278**.

Person int64 43 - 815	Betweenness Ce... 0.053962400288...	Num Crimes int64 8 - 27	Total Co-offenders 41 - 179	Avg Co-offender... 1.889 - 14.6
815	0.1936752154	27	51	1.889
95	0.1171307228	17	179	10.529
43	0.1074956832	8	115	14.375
695	0.09801312292	19	138	7.263
425	0.0766447977	16	173	10.812
715	0.06930401614	19	62	3.263
428	0.06232161312	15	110	7.333
592	0.05951977658	14	41	2.929
178	0.05889090603	21	163	7.762
278	0.05396240029	10	146	14.6

Eigenvector Centrality

- **What** is it? Nodes are ranked according to the centrality of the neighbors. In our case, we will find criminals that appear in highly connected clusters, that could be gangs.
- **Why** use it? We will try to look for groups that operate together among crimes and potential leaders.
- **Issue**: This measure is the most sensible to the projection's artefacts, as high scores will be assigned to large group crimes, and everyone in that clique will result important.

Eigenvector Centrality

- From the table is evident that top 10 criminals have high averages of co-offenders, thus spotting cliques that could be gang and raise the value of centrality.
- As most of the results are inflated by the projection, so they are biased information, but it could highlight the members of criminal groups leaders like **285**.

Person int64 273 - 362	Eigenvector Cent... 0.1717665833876...	Num Crimes int64 6 - 23	Total Co-offenders 95 - 161	Avg Co-offender... 7.0 - 15.833
285	0.3320993759	23	161	7
281	0.3026133914	17	139	8.176
286	0.3015473026	17	145	8.529
283	0.2989114161	16	137	8.562
273	0.2934919978	14	135	9.643
276	0.2934919978	14	135	9.643
278	0.2021218277	10	146	14.6
280	0.1779293757	9	108	12
287	0.1741414235	7	108	15.429
362	0.1717665834	6	95	15.833

Conclusions

- Degree Centrality: highlighted a dangerous crime repeater and many members of big criminal groups.
- Closeness Centrality: highlighted potentially well informed and connected suspects from big criminal groups, even though the risk of artifact is high.
- Betweenness Centrality: highlighted some possible key figures in the information flow of the criminal network.
- Eigenvector Centrality: allowed to spot gangs, but also a potential leader or informant that possess contacts with other important suspects.

Teamwork:

- Divided the work based on our strengths and supported each other to overcome analytical challenges.
- Relied on tools like Deepnote and Google Slides to work simultaneously on data analysis and visual outputs.
- Prioritized class hours for hands-on collaboration, brainstorming, and idea sharing, while maintaining constant communication via WhatsApp to stay aligned on updates.
- Conducted dedicated review meetings and mock presentations to finalize content, coordinate delivery, and provide constructive feedback to one another.

Thank You